

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering **ASSESSMENT TOOL 1 – Data Interpretation**

Course Code:	ITA05	Course Title:	Computer Vision
CO Assessed:	CO4 – Articulation (combining methods for evaluation) P4	Assessment:	Assessment Tool 1 – Data Interpretation
Weightage:	40% of CIA	Total Marks:	100
CO4 Statement:	<i>Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL5)</i>		
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Section / Batch:	Third year	Date:	19/08/2026

Code of Conduct

I _____ Kaviya V/192421022 _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____ KAVIYA V _____

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data	
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided	
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data	
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions	
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand	

Questions:

26. Feature Detection under Illumination Changes

Results:

Method Normal Low Light

A 90% 60%

B 85% 75%

Compare and evaluate robustness.

27. Video Processing Pipeline Comparison

Results:

Pipeline Accuracy Speed

A 88% Fast

B 92% Moderate

C 95% Slow

Compare and evaluate suitability.

28. Motion Model Evaluation

Results:

Model Realism Speed

A High Low

B Moderate High

Compare and evaluate based on application context.

29. Detection under Occlusion

Results:

Method Accuracy

A 70%

B 85%

C 90%

Compare and evaluate robustness under occlusion.

30. System Performance under Constraints

Results:

Model Accuracy Latency Cost

A 88% 50 ms Low

B 92% 80 ms Moderate

C 95% 120 ms High

Compare and evaluate the best system considering all constraints.

ANSWERS:

26. Feature Detection under Illumination Changes

Given Results:

Method	Normal	Low Light
A	90%	60%
B	85%	75%

Analysis:

Step 1: Compare under normal lighting:

- Method A gives **90% accuracy**.
- Method B gives **85% accuracy**.
- Therefore, Method A performs better under normal lighting.

Step 2: Compare under low-light conditions:

- Method A decreases from 90% to **60%**.
- Method B decreases from 85% to **75%**.
- Method B performs better in low-light conditions.

Step 3: Evaluate robustness:

- Method A decreases by **30 percentage points**.
- Method B decreases by only **10 percentage points**.
- Therefore, Method B is more robust to illumination changes.

Conclusion:

Although **Method A** provides the highest accuracy under normal lighting, **Method B is more suitable when illumination changes**, because its performance remains more stable in low-light conditions.

Best choice for varying illumination: Method B.

27. Video Processing Pipeline Comparison

Given Results:

Pipeline	Accuracy	Speed
A	88%	Fast
B	92%	Moderate
C	95%	Slow

Analysis:

Step 1: Compare accuracy:

- Pipeline A has **88% accuracy**.
- Pipeline B has **92% accuracy**.
- Pipeline C has the highest accuracy of **95%**.

Step 2: Compare speed:

- Pipeline A is **Fast**.
- Pipeline B has **Moderate** speed.
- Pipeline C is **Slow**.

Step 3: Evaluate suitability:

- Pipeline A is suitable when real-time speed is the main requirement.
- Pipeline B provides a good balance between accuracy and speed.
- Pipeline C provides the highest accuracy but may not be suitable for real-time applications because it is slow.

Conclusion: For a **real-time video processing application**, **Pipeline A** is suitable because it is the fastest. However, if both accuracy and speed are important, **Pipeline B is the best balanced choice**, with 92% accuracy and moderate speed.

Best overall balanced choice: Pipeline B.

28. Motion Model Evaluation

Given Results:

Model	Realism	Speed
A	High	Low
B	Moderate	High

Analysis:

Step 1: Compare realism:

- Model A provides **High realism**, meaning it can represent motion more accurately.
- Model B provides **Moderate realism**.

Step 2: Compare speed:

- Model A has **Low speed**.
- Model B has **High speed**.

Step 3: Consider application context:

- If the application requires realistic and accurate motion simulation, Model A is more suitable.
- If the application requires fast processing or real-time motion tracking, Model B is more suitable.

Conclusion:

There is a trade-off between **realism and speed**. **Model A** should be selected for applications where realistic motion is more important, while **Model B** should be selected for real-time applications where speed is the priority.

Best choice depends on the application requirement.

29. Detection under Occlusion

Given Results:

Method	Accuracy
A	70%
B	85%
C	90%

Analysis:

Step 1: Compare accuracy:

- Method A achieves **70% accuracy**.
- Method B achieves **85% accuracy**.
- Method C achieves the highest accuracy of **90%**.

Step 2: Evaluate robustness:

- Occlusion means that part of the object is blocked or hidden.
- A higher detection accuracy indicates better ability to detect objects despite occlusion.
- Method C performs better than both A and B.

Step 3: Compare improvement:

- Method B improves over A by **15 percentage points**.
- Method C improves over A by **20 percentage points**.
- Method C improves over B by **5 percentage points**.

Conclusion

Method C is the most robust method under occlusion, as it achieves the highest accuracy of **90%**. Therefore, it is the most suitable choice when objects may be partially hidden.

Best method: C.

30. System Performance under Constraints

Given Results:

Model	Accuracy	Latency	Cost
A	88%	50ms	Low
B	92%	80ms	Moderate
C	95%	120ms	High

Analysis:

Step 1: Compare accuracy:

- Model A provides **88% accuracy**.
- Model B provides **92% accuracy**.
- Model C provides the highest accuracy of **95%**.

Step 2: Compare latency:

- Model A has the lowest latency of **50 ms**.
- Model B has **80 ms** latency.
- Model C has the highest latency of **120 ms**.

Step 3: Compare cost:

- Model A has **Low cost**.
- Model B has **Moderate cost**.
- Model C has **High cost**.

Step 4: Consider all constraints:

- Model C gives the highest accuracy, but it has high latency and high cost.
- Model A is inexpensive and fast, but its accuracy is the lowest.
- Model B provides **92% accuracy**, moderate latency, and moderate cost.
- Therefore, Model B provides a reasonable balance among all three factors.

Conclusion: Model B is the best overall choice when accuracy, latency, and cost must all be considered. It provides better accuracy than Model A while avoiding the high latency and cost of Model C.

Best overall system: Model B.

